

Collective Behavior (Discussion) ~ Daniel Friedman ~ Active Inference for

Courses/ActiveInferenceForTheSocialSciences | Daniel Friedman ~ Active Inference for the Social Sciences 2023 | 1:21:46

Hello and welcome. It's August 16th, 2023. We're in the collective behavior discussion section of the active inference for the social sciences course in 2023. So perhaps let us all go around and give an opening introduction and or reflection and see who joins and what happens from there. But just so that everybody has said hello.

Hi, everyone. I'm Brun. 9pm in South Africa. And yeah, just excited to be and see what happens.

Hi, I'm Andrew. I'm a graduate student in Indiana. And I guess my field of interest now is the knowledge management. And I think that figuring out how we collectively learn and can share information. It's a really interesting topic. And I'd like to see how this would apply.

Hi, I'm Regina. I'm from Guadalajara, but I'm living in Spain, studying and doing experiments related to this topic and trying to learn a little bit of the theory that's behind it. So we can, you know, advance better. I'm excited.

Hello, my name is Sasha. I'm calling in from California. I'm also a graduate student.

Yeah, just really enjoyed the spirited presentation last week. I'm looking forward to a discussion.

All right. Well, how about anyone can give a first reflection on collective behavior? What was their thought or context before and where are they at as we kind of begin? Yeah, Regina?

Okay, so I just want to make sure that I understand because I took like two bigger lessons from your lesson, right? And the first one is that, you know, this, this famous model of active inference that has the agent and the environment and the mark of blanket in the middle, right? And just interacts with everything, right? So I think what I got from what you discussed, maybe I didn't have a good definition of an agent. But first, but you can have that agent not be just an individual, but also your definition of an individual changes, depending on what you want to study in the behavior, right? So an agent can be an ant, it can be

an ecosystem, it can be an eye looking at something, or it can be as a society, for example. And I think that's a big

take for me because it changes a lot the concept of how to study things and use active inference, just by changing how big the agent can be. So I don't know if I got that right, right? So first that clarification.

So what is an agent?

Yeah, so I don't know if there's like a formal definition of an agent in active inference, right? But if we, if we want to study whatever we want to study, because I'm not going to study because I have to do

experiments. That's like my goal, whatever I'm doing. So I want to study whatever I have to choose what's going to be my

agent and the environment it has. So choosing the agent, you can do it to create a model or to make an experiment, whatever. But the choice you make, it will depend on your, what you understand as an individual or as an agent, if I think so, right?

Awesome. Anyone have a thought on that?

No. If I continue with just to add something else and this will be my, my, um, and then I'm thinking is that, um, also a big problem that we have, or I think I have is that since I am part of the environment, um, you know, you know, the perspective of whoever's studying whatever it is, it's also going to be an effect in it. So maybe, maybe scientists already decided, but I don't know that we can never have an objective way of studying something if you're part of the environment of the agent. So I don't know, we already forgot it or we already decided that's it. We cannot have objective observations of behavior in general, but you know, you always think about like the agent environment and you studying it as something else, but there's something else is also affecting it. So I see that as a problem or maybe something that has to be said, like we are not objective because we're part of what it's affecting our agents in the behavior collectively too. So those are the thoughts I have to share.

Uh, Regina, I'm, uh, I'm also new to, to all of this. So my preliminary understanding of, of what is an agent is, uh, anything that has any good reason, uh, to self evidence and to, uh, itself or, or to survive, um, that doesn't want to dissipate that might delineate the boundary of what an agent is that then sets up, uh, um, uh, the boundaries with which it, um, will, as you described, start interacting with the, with the world, with the environment. So if something, as I understand it, if something has a reason to go on being, to survive, um, um, then it is an agent in the world.

Okay. But, but that agent could be anything, right? Like, is that what you're saying, right? Like if he wants to survive, it can be an ant surviving individually or a cell or, or a group. Okay. Yeah. That's a good, that's like a good definition. Okay.

Yeah. Sasha?

Um, great points, Regina. Um, so much to say there. Um, it's something that I have experience with, um, as a researcher in biology is deciding the kind of, um, the statistics or sampling approach to use, whether we're taking, you know, one cell from one individual, or we're taking many cells from many individuals and how to do the statistics and kind of, um, sampling approach on that. Um, that is a, you know, an evergreen question that we're always reevaluating to see, um, how to think about the independencies within biological systems. Um, and then another thing that you mentioned was, um, about the individual being part of the ecosystem and that makes me think a lot about complexity science and something that, um, is addressed there in, uh, thinking about the interplay of the scientists and what they're studying. Yeah. Thank you for those points.

Uh, Sasha can, sorry, I'm not raising my hand. I'm just unmuting if that's okay. Can, can you say a bit more about what you mean by complexity science? Is.

Sasha Tkacheff-

Maybe muted.

Oh, can't speak at the moment.

Yeah. Complexity science has looked at agent, agent-based modeling as a fundamental tool.

John Holland signals and boundaries focusing on what from an active inference side, we might say

But there's a lot more to say in that. Who wants to bring in some new angle or how do they see collective behavior? Why do we talk about collective behavior?

Andrew, anyone have a thought on that?

I'll bring in a definition of umwelt and then we can see.

But meanwhile, does anyone have a thought?

Daniel, we don't know if you're talking.

Sorry, I was just muted on Zoom.

Andrew, you're a little quiet,

but I'm going to bring in a definition of umwelt.

And then meanwhile, does anyone have a thought?

I hope I'm not

further diluting Andrew's question.

But I think of collective behavior in kind of like a,

I don't know, a football team or a soccer team

and what that provides

the group who is, uh, shares some common ethos

and what that does to the, how that regulates the,

their collective or individual and collective free energies,

um, during a match or throughout the season

and how that gives some, uh, yeah, some charge to, uh, some investments

because it's my team against yours and we have to, um, try and win.

And so, uh, you know, without, without that collective behavior, um,

it's, it might be far harder to achieve

a group's goals if they aren't able to collaborate and share,

um, you know, implicitly and explicitly

certain traits, behaviors, thoughts, feelings, ideals, ideologies,

whatever they might be,

and how that then allows us to compete in the world.

I, I, I see what you're saying.

It's, it's brilliant?

Yes.

Okay. Okay.

I see what you're saying.

Um, especially because we're watching the world, the world cup, right?

Women's world cup.

Right.

Right.

Yes.

Yes.

So it's very relevant.

[illegible]

[illegible]

Here's another question.

So what is the role of any technical detail in the social sciences?

Whether more on the statistical, connecting with data, or more category-theoretic-like, just formal?

What's the role of formal methods in different kinds of academic or practical expressions about social systems?

Does it matter at all how we think about a framework at all?

Well, for formalizing aspects of a social system, I would say that the purpose is for communication between individuals.

Because just, you know, we can't verify that, well, formalizing it is just setting a system on top of what you're observing.

But there is no explicit guarantee that what you're saying encompasses everything of the phenomena you're observing.

And just, like, you know, kind of going back to Taoism of, like, the name that can be named is not the true eternal name.

And so it's just a way of, you know, it's a slice of what we can observe.

And the way to, you know, maybe you can make predictions or maybe you can generate something that acts like the phenomena you're observing.

But, so it's just, it's a way to kind of...

It's just a way to kind of approximate our understanding of some phenomena.

And you have to slice it in a specific way.

But also understanding that it is a slice or, you know, it is a tangent line that you're approximating as opposed to encompassing the whole system from verbalizing it.

So, let's make this

Kinda. So you write the instructions and try to see if you can outsmart someone else.

I mean, it sounds kind of dark, right? But maybe it's one of the goals too, or something that just happens out of knowing the rules that nobody else knows.

I mean, I guess when you talk about control, there is a shade that can be gained from that specific wording.

But control in terms of control theory, or that comes from electrical engineering, it's really just what shapes the behavior, or what are the inputs that drive the behavior?

And when you're talking about control, it's really just about, well, if you change these inputs, how does that change the end behavior?

And so you can actually, from electrical engineering of controlling some sort of electromechanical system, but

then you can make analogous statements on control into society, which doesn't necessarily mean the dark negative control.

But like, you know, maybe there's some pathological behavior in a society, or, you know, even pathological responses within the brain.

And it's like, well, what is driving that? And is it possible to change the drivers of that to have something that is more beneficial to the system?

Yeah, just to add to what Andrew is saying, because it matters sometimes how you manipulate, you know, a system, or it matters how you affect a brain.

So it's not just kind of poking and prodding.

I mean, you know, we've come a long way, for example, from lobotomy, which had disastrous effects.

It matters to a system how you affect it.

And so to know more formally the most effective way to create a desired effect, whatever that might be, it's essential that it's not just kind of poking and prodding.

As much as that is part of the scientific method, to update that method in accordance with what you're learning is important.

No, I think you're both right. It doesn't have to be dark.

But also something else that Andrew said about predicting, I think that's one of the big goals.

Trying to predict whether to be prepared to, or just to, I mean, it couldn't be to control, right?

But if you can somehow predict how people respond to data or how a system will behave in this other scenario,

you can, you know, change what the input is or prepare for whatever comes, I guess.

Yeah.

A lot of great points.

Some of the commonly held, not equally by everyone, everyone, but some of the tenants of science being to explain, predict, design, and control,

where explanation and prediction are on the pathway towards systems designing and maintenance and other words that people can add.

But that's the pragmatic, utility oriented view, which doesn't have to be an absolutist view.

But certainly there are regimes of attention socially now and seemingly for long, long times where people were wanting to retain and to constrain the variance distribution on what we would say is their generative model, some norm.

And then other times engaging in these functions and that instrumentalist usage of active inference is like, we're going to be able to model it with a pretty minimum framework that's commonly used in statistical systems.

where some things we observe and some we don't.

And at least we know and how all the pieces are connected.

And then that is an artifact in model based design.

[illegible]

[illegible]

Like there's a paper in here, here, because we create the knowledge and we tell everyone the knowledge.

This is the model.

This is how everything behaves.

And then we update our knowledge about it.

And then we look for everything to just, you know, confirm it.

And, and yeah, maybe there's other system that will not let us be getting knowledge.

So slowly, because if everybody just agrees on something, we teach it to everyone and we confirm the model.

And then we just try to look for evidence for that model.

Then we go slowly if there's any better thing or better model that comes along.

Yeah, a lot to say about that.

Like if we believe that trees exist or that people exist,
and then we come to see them.

So there's kind of this sense where problem confirmation bias, which I think is a great term, a great topic to bring in with definitely also what's called like the golden nail syndrome or something.

If all you have is the hammer, then everything's a nail.

And there's definitely attention there because it's like, how much of what you can do are you supposed to do?

Something that you can't do?

Or, but then at least you could do what you needed to do to get there.

So that's actually within your scope of what you can do.

So it's not that there are things that you can't do that you can't do.

If just from a measurement perspective.

So like, how does it look?

Or how do we recognize across systems?

Or how do we have awareness in situations as limited individuals who do have all of these cognitive phenomena, how to make decisions or sense-making or infrastructure

or whatever it is, the socialness,
the system of interest of the social sciences,
make that and then dot, dot, dot normatively what?
I think it's, there's some protection built into,
into the problem of confirmation bias
because all knowledge generation is happening in an open system.
And so there's always going to be an agent outside of your,
your own mind and it's, and what it's perceiving
that can then challenge or add to the,
you know, like the, the, the super,
the, the LK99
semiconductors that have,
has been all the rage recently
and how other scientists have now kind of
discredited it quite convincingly.
And it's because of this open,
they might've had degrees of confirmation bias
that they were egging each other on,
but it's, it's ultimately been refuted,
which is this built-in kind of protection
because it's all operating in a, in a,
in an open system, fortunately.
Yeah, a lot to say there.
Monday morning reviewer two-ing.
How, how do we, again,
go from any given description
of a communicative event,
like the kind of specific unfolding
of the LK99 narrative
as yet uncompleted
in, in the minds of some,
which is real enough
to be real socially.
And so how do we go from any descriptive,
anything about that
total observation
to core screening?
Then how do we

come to work with that interface
with what, whatever that is,
hands that to us?

What is the social behavior?

What are some, some classic
or some novel social behaviors?

Maybe ones that seem pretty obvious
due to recency bias.

I'll, I'll just mention one aspect
that Andrew and I have also talked about
is the autopoietic nature
of information systems.

Andrew, do you have any thought on this
or about where autopoiesis
comes into like social sciences
or these kinds of models?

Daniel, Andrew just left a note
that he's, uh,
needs to separate away momentarily.
So he'll be back.

Oh, totally.

He just wasn't looking at the time.

Funny stuff.

Anyone else have a thought on that?

Okay.

Okay.

Just, just to kind of
carry it forward then
basically the, the ability
of social systems
that include
written niche modification,
cumulative culture,
their increasing
extent of control.
as discussed earlier,
that definitely is a bit of a
overloaded or at least doubly meant

term in different settings
because it can just simply mean
modeling it with control theory,
which is pretty limited philosophical
statements relative to any
statistical method
or control.

Like the kind that people
are unhappy with
and the kind that's described
in thinking like a state
and a lot of other works
that I know

Mao and Lorena
and Abel
are going to be
definitely
going into more
detail
on.

So when societies
can model
and introspect
or, or
self-reflect
formally
much better
and that can happen
through
anything from
free and open source
better rated protocols.

There's many limited formats
of self-meaning making,
but cryptography definitely
allows those explorations
and that doesn't necessarily
mean computer cryptography.

That could also be like
linguistic encoding
and all these other
ways of,
of niche
sub-differentiation
in multi-scale systems.

You just
do not see
a system
that is
cookie cutter.

It's very
rare
in ecosystems.

There are
algal blooms
and things like that.

So when we really
think about the
developmental
and the cognitive
and the selective
niche,
which is a
concept that
Axel Constant
and others have
differentiated in some
of their like
ecological
active inference
work,
like that
niche
is very
embodied,
extended,

and cultured,
et cetera,
et cetera.
It's just like a kind
of cool,
direct way to
open up
to that challenge
of modeling
and just
bring some
closures
with different
accounts or
compatibilities with
different accounts.
So like
all,
just to give one
random example,
but maybe it'd be
interesting what other
people think,
all the expressions
that people make
in social settings
can just simply
be their expressions
or modeled as
such.
That is
in one way
implicitly
what was always
in play,
but now
it's
describable

and legible
and autonomous.
So
how do we
take this
active inference
collective behavior
crossover
and think about
what it really
means
in how
we're going to be
like thinking
differently,
even just this
discussion forward
or like
more locally
our
understandings
with collective
behavior
or with behavior.
Daniel,
can you ask
the question
again?
I didn't
actually catch
the question.
How do we
start to adapt
some of these
general
comments
or frames
that are

in some ways
disconnected
from descriptions
of real systems?

how do we
think about
that
and what
it means
now?

What is it?

Do you mean

on the
document
in front
of us?

These
disconnected
things in front
of us

on the
document?

I don't know.

Whatever you think
or whatever
you want
to say.

Okay.

Yeah,

I think

there's a lot
and again,

I hope I'm not
fully understanding
the question.

I hope I'm not
steering away
too much,

but I've got
a general sense
of what I think
you were saying
and
I think
it's an
enormous
problem
with the
amount
of
information
sharing
just in a
with
let's just
take
the
generative
models
of
social
media
and all
of the
platforms
that are
available
to us
how much
information
we are
privity to
in any
given
moment.
I'm

kind of
forgetting
the
question
that you
asked,
but how
do you
it's almost
as if
there is
such a
high rate
of information
and data
flow
that it's
really hard
to know
what to
do with
all of
it.

At least
as a
human being,
I'm not
saying for
machines,
but as
a
psychologist
I'm
certainly
interested
in how
humans
are able

to
tolerate
and
respond
to
the
amount
and rate
of
information
that's
out there.
the
time
that's
happening.
This
is not
the first
time this
happens,
right?
Like,
we went
from talking
to writing
information,
then everyone
wrote their
information,
and then we
have books,
and then we
have libraries,
and that's
all available,
and again,
we have access,

if you can
read,
right,
to all the
information
in the world,
also,
we adapted
to,
I don't
know,
to get
careers,
so we
could specify
which type
of information
you are going
to know
more,
so maybe
we need
some sort
of new
career
types,
I don't
know,
more
specification,
with more
information.
But is it
not,
is the rate
of information
flow and
increases,

is it not
exponential,
is it not,
it's a
crazy amount
and our brains
cannot do
anything about
it,
so we have
to do
multidisciplinary
things and
work together
with others
if we want
to make
something,
but,
yeah,
I don't
know,
maybe we
will die
of excess
of information.
how do
people feel
like active
inference
brings something
to the
discussion
relative to
how they
want to
work
towards?

Ants?

Yeah,

well,

isn't that

the provocative

question,

too?

But like

any system,

that's the

question.

Everyone is

going to

have their

own memes

and themes

and systems

that they

have spent

time with

and did

fieldwork

with and

all this

other

particularity

because a

system that

doesn't have

particularity

is just

not going

to be

hosting a

broad enough

niche for

phenotypic

diversity.

It's just
riding on the
edge of a
razor instead
of more
like a
fabric that
continues to
unfold,
I think.
there's a
lot of
work to
do with
the
social
sciences
and
active
inference
just from
what I
was
like
reading,
trying to
find things
for this
discussion
and just
I personally
wondered what
is meaningful
or interesting
or useful,
what's
epistemic
and pragmatic

value for
different
audiences?
How does
it go from
kind of
bringing
citations that
were not
linked
together
in
some
more
theoretical
or more
generative
model-oriented
work in
active
inference
like
epistemic
foraging
or the
inference
model
or
any other
kind of
specific
model?
What do
these
different
kinds
of
scientific

artifacts
do
for who?
So,
what do
I guess
I'm still
learning
very
crudely
plodding
along
here,
Daniel,
to try
and answer
that because
I've quickly
realized that
the field
is so
multinodal.
It has
so much
the
free
energy
principle,
the
Bayesian
brain,
even just
to psychology
has so
much
applicability
and
everyone

I meet
at these
discussions
is coming
at it
from a
different
angle
and it
can be a
bit confusing
to hear
how other
people
are
applying
it,
how other
people
are
making
use of
it in
all
manner of
ways.
And so
I'm still
learning
how to
apply
to kind
of ring
fence
the free
energy
principle
and connect

it to
psychological
processes,
including
collective
behavior,
because there's
a lot of
psychopathology
in group
thinking in
collective
behavior.

But I'm
also very,
very interested
in its
application at
the interface
of what
happens
between a
therapist
and a
patient.

I'm often
unlike other
professions,
who seem
who are just
by nature
of the
scientific
method,
are able
to extract
a lot
more data

to work
with,
to analyze,
that's
very hard
to do.
I feel
very limited
other than
just making
kind of
theoretical
leaps and
inferences
about how
to apply
the free
energy
principle
to psychology.

I can't
have a
session
with my
brain
hooked up
to an
fMRI
machine to
get data
and to
see when
a patient
says,
boo,
go,
bar,
that's a

spike on
the spike
trains,
which represents
surprise,
and we can
then solve
for that in
some way.

I don't have
access to that
sort of data
flow, so
it feels
limited, but
I'm still
learning how
to apply
it and
trying to
collaborate
with
colleagues
on the
inference
group.

What are
people
wondering
about
collective
behavior?

Even if
we're totally
of course
not going
to address
it, but

just what
are the
kinds of
things that
people...
What makes
you look up
collective
behavior
instead of
just behavior
or individual
behavior?
For me,
it's about
safety and
need to
belong.
people
to
people
and
people

are
in the
world.
For me,
it's...
I think
that it
depends on
where you
come from
and the
country you
are, but
there are

places that
are more
collective
and there
are places
that are
more
individualistic.
and I
think mostly
Western
countries have
a lot of
collective
behavior and
knowledge and
spirits already,
right?
That they're
already new,
these things
that we
think are
new for
us, right?
How the
collective
thing works.
For example,
I don't know,
you don't make a
decision for
yourself,
you make a
decision for
the group,
like marrying.
you marry

because you
want an
addition,
you want to
interact with
someone or
other group,
and that's
something that
affects the
collective,
not just
you.

So,
arranged marriage
makes sense,
something like
that, right?

So, for me,
making that
distinction about
individual
collective comes
from understanding
that, I don't
know, there are
other places
that are already
ahead in what
collective means
in terms of
human beings
and how to
interact.

and the
Western world,
we have all
this individualistic

thing, which
is very
interesting because
why do we
have the same
species and
then we have
very different
approaches of
how to
understand
ourselves in
terms of what
an individual
is and why
it matters.

Yes, there's
so much
loaded on
different
understandings
among people,
including even
the essence
of the
social, which
is who are
people and
what is the
obligation and
how do we
view that
relationship.
relationship.
That's a
very deep
area and

again, the
part that
just kept
returning to
me in the
lecture and
when I was
trying to
think about
how to
even reflect
part of
this very
large area
is like,
what is the
adjective
doing?
And what's
the reference
point by which
this is being
said in the
speech act?
like people
often will
dismiss, I
don't mean to
you know,
straw people,
but dismiss
collective
behavior by
saying, well,
we are
explaining
individual
behavior.

What are you
explaining beyond
the nest
mates in the
colony?

And then the
exterior, like
that is it.

And so that
can feel like
looking at the
same thing from
two different
sides.

And there are
different methods,
like some methods
blur over
individuals.

a lot more,
like statistical
methods that
might correspond
to demographics.

And then on
the other hand,
like more
agent-based
models.

But how
much
variability and
what shared
variability
and what
general
synchronies
amongst

different quantum
reference frames,
all these
nested,
at least
possibilities,
possibilities
that are
grounded in
wanting to
explore
mathematical and
scientific
possibilities,
as the
literature of
linear regression
is full of
so many
incredible
extensions
and similar.
things like

And then
that's,
I think,
what the
course is
even beginning
to try
to express,
and what I
understand in
the works of
the other
teachers and
like a lot of

your backgrounds
and everything
like that,
is where
is that
moment where
it is not
only science
because science
is not
simply the
social.

So science
as an
activity,
and then as
there's a
diffusion away
from science,
and I think
that,
or at least
that relationship
between social
and science
is right there
in the title.

Maybe that's
kind of like a
science major's
take,
but it is
itself in a
way,
presenting a
dilemma
with what
science means

there,
and then that's
the frame game,
who's doing
social science,
who's doing
not social
science,
and that's
where,
in the
culture I'm
sitting in,
slash,
in the technical
side of that,
it is related to
statistics and
formality.
It's the
ex-KCD
comic,
and the
way that
formalisms
and details
are taken
as,
under some
regimes of
attention,
basically definitive.
And yet,
there's like a
yearning and a
counterbalancing
that's expressed
also in many

ways,
and so that's,
and science,
maybe some
people feel like
certain social
scientific methods,
that's like,
not far enough
in this dimension,
and then the
others feel like
they're too far
in this direction,
but it's like,
just like any,
like that
academic field,
let alone any
other community
of diverse
agents,
like,
of course,
there's going
to be
distributions
of perspectives
at different
times.
And,
and uptakes,
rate of uptake
by those
different perspectives,
and,
you know,
people.

Yeah.

Daniel,

do you think

these first

principles

would be able

to explain

why an ant

colony exists?

I know that

was your,

kind of,

area of interest

of study.

Why an ant

colony exists

in the way

that it does

as a collective,

operating as a

collective,

and why

another

animal

is able

to survive

and achieve

its goals

as an

individual,

only relying

on the

environment

and its

actions and

sensations

to get its

needs met

and to,
no doubt,
minimize its
free energy.

But can
first principles
explain
why the
one has to
survive as a
group and
the other
has to survive
or is able
to survive
under different
conditions?

Yeah,
that's a
super interesting
question.

It's very
core to
this
individuation
and
multiscale
individuation
concept.

Like,
is the
nestmate
the individual
and then
the colony
is the
social?

Because then

for sure
it's super
collaborative.

Or is the
colony the
organism and
nestmates are
tissues?

And then are
they social
across colony
boundaries?

Not where I
was in
Arizona.

But they're
often friendly
across different
species and
they avoided
conflict more
often than
not.

So it's
like,
why does it
even matter
what's happening
on this
plot of
land in
this area
from a
biomimicry
perspective?
we might
learn new
algorithms.

It's a
sandbox for
learning new
techniques and
observation,
all these other
richnesses.

So it's to
say,
what comes
away from
that science
that makes
it social
science?
things like

And I
think there's
a lot of
disciplinary ways
of understanding
that, which
I'm definitely
not expert
in, in the
professional
fields like
counseling,
but probably
in all the
different
humanities and
histories and
all these other
fields,
all these
different maps,

the kind
of social,
the epistemic
organizing
question,
and then the
social organizing
question,
and the way
technology is
also in that
tangle.

And yeah,
someone pointed
out like this
is not a
new problem,
but then it's
like, well,
what parts are
new?

And then
what's our
model by which
we're saying
that it's
like new
or too
individualist
or too
collective
on what
scale?

What if
they're like
really
collectivist at
this scale

but not at
this scale?

Or an
example of
an ambiguous
or at

least like
polys,
polys,
phenotype of
colonies is in
some colonies
there's partial
reproductive
activation of
like non
reproductive
nest mates.

And that
has often
been modeled
from like a
position of
like zero

is the
law and
how are
they cheating
above this?

But it's also
been approached
from a
colony level
strategic bed
hedging.

So without
super deep
knowledge of

really some
of the
specifics of
the regularities
that that
colony algorithm
had to persist
for so long
and to have
gotten swept
off the table
if it was even
it's like
that's a
very
integrated
problem.

So it
helps you
approach
collective
systems and
learn about
them but
it's like
there's a lot
to say about
humidity loss
in foraging
and humidity
loss isn't even
the whole
picture.
and human
foraging is a
lot more
sophisticated
from just

like a
perception
cognition
influence
perspective.

so those
are just
some ant
related
thoughts.

But it
just centers
a lot of
discussions
that really
matter.

And I
think the
really
interesting
piece is
not always
in the
way that
they're
thought to
by some.

Yeah,
and it's
very hard
to get
any kind
of cross
pollination
or to see
cross
pollination
studying one

agent.

It's quite,
or at least
in my mind,
very difficult
to kind of
generalize
things across
different
agents.

It starts
getting a
bit,
yeah,
tricky to
infer one
collective's
how and
why and
how they
survive and
how another
agent does.

It's tricky.

So I'm
almost just
going to
make an
agent,
human
beings as
hard as
they are
to study
and just
try and
run with
it as

far as
possible.

Yeah.

Otherwise,
it's just
information
overload.

Body's a
complex
place too.

And
digital
technology,
virtuality,
these are
such
richly
developed
topics in
the social
sciences.

I only
know the
tiniest
amount
from,
but I
think that
through
participation
and inclusion
in collaboration
and in
conversations
and everything,
there's going
to be a lot
of interesting

connections

paths.

And it's

being approached

seemingly

multiple

layers,

like within

the two

layers that

just we can

focus most

narrowly on.

It's like

within the

realm of

science,

what is

science,

which is

kind of like

playing this

other role

with inertia

in some

people's minds

and in

systems?

What is

science?

Who gets

to decide

what social

science is?

I don't

know.

I'm not

in social

science,
but it's
systems of
interest to
society and
certainly I'm
a part of
that.

So it's
like,
what is
good
technically?

What is
social
scientific?

And then
what is that
social scientific?

How is that
going to
interface into
the metaphysical
questions and
the normative
ethical and
the decision
making with
trade-offs and
the real
reflexive,
complex
decisions that
matter also
while recognizing
this increasingly
nuanced way of
thinking about

different social
intelligences?

what's any

other random

thought or

curiosity about

collective

behavior?

behavior?

One point

that I

would have

added in to

the lecture

was just

that in

distributed

systems,

whether as

modeled or

as such,

all behavior

is arising

from interactions

local to

individuals.

Even if it's

sent through

some wireless

mechanism,

it's like all

about the

local

interactions.

And so

that is

an arising

type process

that gives
the modeler
one hands-off
layer at
least,
if not much
more.

Another nice
feature about
them,
they are
in a way
waiting to
be composed
because they
intrinsically
contain at
least two
layers of
consideration.

now whether
there's a
forcing function
of one
layer and
dangling a
bottom layer
or whether
there's a
lot of
emergence or
discussive in
that way,
that more
arising function
can be
modeled.

But in

either way,
you can wrap
that two-layer
system and
then compose
them.

So that's
what enables
in certain
situations like
parallelization.

Or at least,
you know,
that's obviously
a big area,
but that's
what allows
multi-agent
simulation
frameworks to
do well
on certain
very challenging
sense-making
and action
selection
problems,
not by
guarantee or
default,
because any
given slicing
and dicing of
any given
prompting of
information may
not help you
adapt.

But there
are paths
that do and
don't help
persist or
not.

And that's
this kind of
persistence
starting place
that we
began with.

Sasha?

Yeah, I
really
resonate with
what you
mentioned about
the local
interactions
that even
if information
is transmitted
over long
distances,
then the
agents
behave in
a way that
is locally
relevant and
integrates that
information.

And it
actually makes
me think a
lot about
the psychology

therapy angle
that while
you try to
get information
about a
person's
sort of local
environment and
behavior,
or like
perhaps demographics
and age,
that their
informational
environment and
wealth of
what kind of
information they
consume might
also be
really telling
about their
psychological
behavior interactions
or pathologies.
And then to
bring it kind
of full circle
back to that
the researcher
is part of
the ecosystem
that they are
studying.
thinking.
Yeah, I think
it's just
such a

difference in
perspective to
see that as
a positive and
a gift,
that the way
that you would
design and run
an experiment
today is not
the way that you
would do it two
years from now
with more
experience.

And so
that is a
benefit and
like a bonus
in the system
and not
like a negative
perspective of
bias that it's
detracting from
some underlying
truth or
reality.

So just a few
thoughts.
very interesting.

Anyone want to
add something?

I think we
didn't fully
resolve this
early question
about the

umwelt
term from
ethology,
behavioral
research
observation
science,
using the
term to
describe the
world as
it's experienced
by a particular
organism.

I think a few
different pieces
and there's a
lot of discussion
in discourse
or just to
just give it
like a super
short overview.

The world,
we might connect
to world
modeling and
there's a lot
of interesting
work in all
kinds of
different like
pros focus
on qualitative
systems as
well as some
artificial or
synthetic

intelligence,
world modeling
as it
experienced,
as it is
experienced is
one of the
sticky and
deep ones
because this
is the
area of
phenomenology.

And so
this is like
a kind of
semi-eternal
discussion point.

So whether
active inference
delivers on
world modeling
is kind of
empirical and
just plain,
whereas as it
is experienced
is going to be a
philosophical
question.

And then by
particular
organism,
that specificity
we get by
actually making
the generative
model,

even if we're
just speculating
about it.
It's like
talking about
experience without
an experiencer
would be like
talking about
some agent's
behavior without
reference to
its niche.
event where
there is like
that complex
negotiation.
I think the
humanities and
a lot of the
social sciences
have identified
a lot of
the
structural
top-down
and also the
personal
bottom-up
motifs of
things being
blurred and
made complex.
and so I
think part of
the social
and the
scientific,

as coming
only from
the scientific
side,
is like to
meet in the
third space
and or
at least
make it
work to
have a
mutual
compromise
where it's
like certain
social-oriented
claims based
upon this
scientific
measure that
is going to
be a judgment
call,
but understanding
that,
conversation,
where is it
a mutual
compromise
in like
what
dimensions,
not simply
along the
kind of like
rigor or
accessibility

or implications

axis,

there's just

very many

things in

play in

these

iterated

relationships

in

ecosystems.

having a

researcher as

part of the

ecosystem,

well,

here's the

forager question.

You see the

nest made on the

ground, it's a

forager, but

then we step

back, it's like,

okay, well,

it's foraging,

and then that's

the allocation

of the forager,

then its

foraging is

still part of

this sense

making, this

generative sense

making with

like process

understanding,

even of a
trace, which
is something that
Bruno Latour and
others have
explored with
like following a
trace in a
social setting
and using that
to trace
networks.

So in our
digital ecosystems
though, our
information
ecosystems, is
not research
happening?

And so what
does that really
mean?

And then what
are those social
normativities?

There's going to
be like a
meeting or a
handoff with
where that
really is
happening.

Or maybe it's
a knocking on
the door that's
ignored.

Or,
it could be

whatever metaphor
in that setting
it actually is.
There's no
general metaphor
because that'd be
like, oh, what's
the general metaphor
on this experiment?
Which experiment?
Just the general
action that we
plan to take?
Oh.
Hard to say.
Okay.
Okay.
Well, in our last
like,
actions, I guess.
If anybody
types of question
live chat, we'll
look at it.
Otherwise, what are
any other
thoughts or
like...
What?
What?
Can we learn
more about
collective
behavior?
Or what will
it mean for
our sense-making,
decision-making,

if we understand
collective behavior
better?
we'll

known to

Thank you.

Thank you.

Thank you.

Like based upon the sections that we've done so far, and how we can even start to approach this in an acknowledged, limited, finite way.

Yeah, Sasha, then anyone else?

Should we embody in order to improve cohesion in collective behavior?

It makes me think of certain psychology traits or descriptions of like openness or conscientiousness.

What will make individual humans better participants in collective behavior?

Yeah, I think to add to that, to just emphasize the importance, again, only with human agents, specifically, who have the capacity to think and reflect.

And the only ones that have those really advanced capabilities that we know of as yet.

But to really kind of leverage in order to make the collective better, there's a responsibility of the individual, despite being part of a collective on many different fronts,

to be able to leverage and utilize their own internal states and what they have learned and to be able to think in a, to retain the individual, the individual's mind and think about the collective.

I think it's so important to, I think it's so important to, to become healthier, Sasha, to your point and to improve its, yeah, to become more conscientious, to become more responsible or whatever that might be.

So to always try and maintain a thinking self is crucial.

And once you, you can very easily be usurped into the collective as it is.

If you just go on any social media, media platform, you will see individuals rarely thinking for themselves and seemingly kind of just fueling each other and that, that causes massive problems.

So, an individual mind is, strikes me as being just the most important thing in the future going forward.

Yeah, this dialectic with individuation and collectivization as these like ongoing negotiated compromises that can be also viewed many ways, however, however they are.

Cognitive science coming into play.

Maybe historical processes or philosophies were used in many settings for even approaching these kinds of social questions.

And now with computer science and technology.

And even like is a discussion more about the capacities of measurement and action.

Not that those are new in and of themselves, like, but thinking like a state, like the development of the apparatus, the development of the social social systems that can be run on, on open source hardware, open source software, things like that.

So, I don't know if something's generalized across all such groups.

But understanding what environments, at least initially things are beginning at, is just a realist starting point.

Probably no guarantee since most things slash all things are swept off the table in their specifics anyway.

So, and how to deal with that kind of like skin on the line type of decision making is very interesting how it's pursued in the sciences.

Because often like many runs are presented or that kind of like catastrophic failure mode of models is not considered just like part of the training starting naive.

So, just the way that different social considerations are converged upon.

It's never, it's not like reduced to one feature of the social because the social is such an open space.

But I'm always impressed by the students.

I'm always impressed by how that writing could kind of open up the situations which are like mundane, but also of course very deep and meaningful.

Also though mundane.

Thank you.

Thank you.

Thank you all for participating.

David, want to add anything?

Thanks, Tim.

Thank you.

Thank you.

Thank you.

Bye.

Bye, Bruno.

Ciao.

Bye, Bruno.

Thank you.